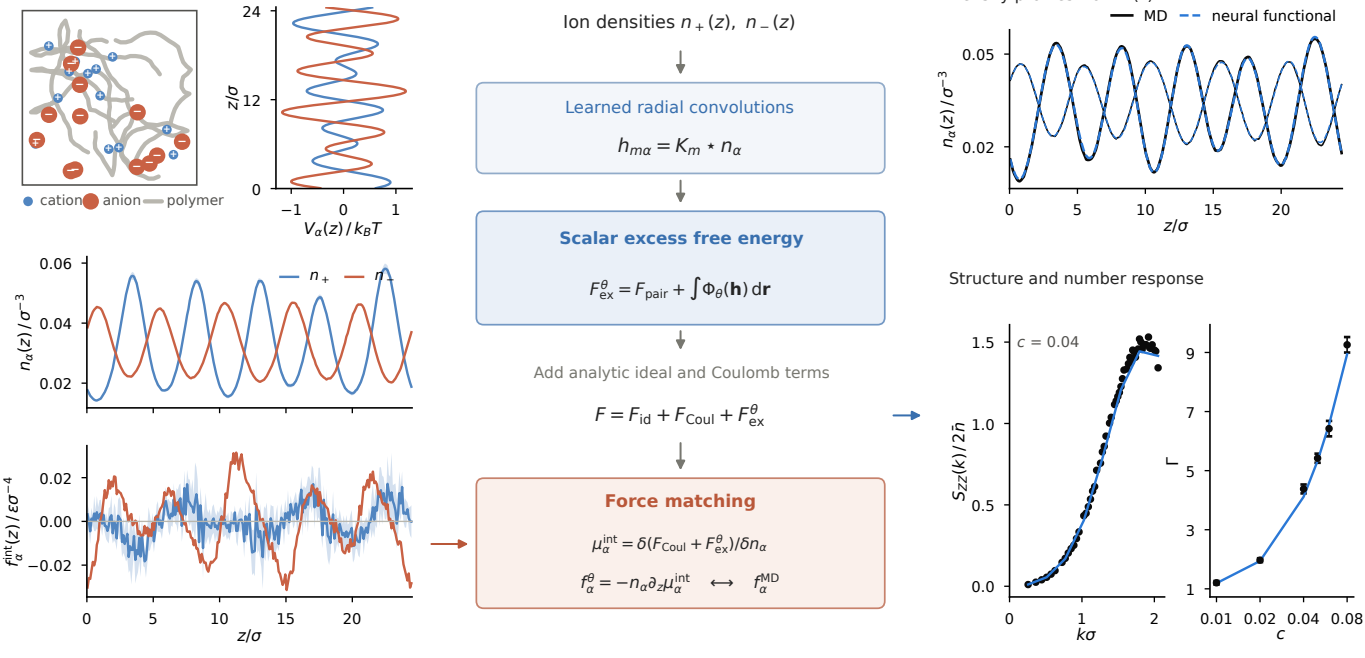

A Simulation data
cation
anion
polymer
24
12
0
z/σ
−1
0
1
$V_\alpha(z)/k_BT$
0.06
0.04
0.02
$n_\alpha(z)/\sigma^{-3}$
n_+
n_-
0.02
0.00
−0.02
$f_\alpha^{\mathrm{int}}(z)/\varepsilon\sigma^{-4}$
0
5
10
15
20
z/σ
B Functional and training
Ion densities $n_+(z),\ n_-(z)$
Learned radial convolutions
$h_{m\alpha} = K_m \star n_\alpha$
Scalar excess free energy
$F_{\mathrm{ex}}^\theta = F_{\mathrm{pair}} + \int \Phi_\theta(\mathbf{h})\,\mathrm{d}\mathbf{r}$
Add analytic ideal and Coulomb terms
$F = F_{\mathrm{id}} + F_{\mathrm{Coul}} + F_{\mathrm{ex}}^\theta$
Force matching
$\mu_\alpha^{\mathrm{int}} = \delta(F_{\mathrm{Coul}} + F_{\mathrm{ex}}^\theta)/\delta n_\alpha$
$f_\alpha^\theta = -n_\alpha \partial_z \mu_\alpha^{\mathrm{int}} \quad \longleftrightarrow \quad f_\alpha^{\mathrm{MD}}$
C Predictions
Density profiles from $V(z)$
MD
neural functional
0.05
0.02
$n_\alpha(z)/\sigma^{-3}$
0
5
10
15
20
z/σ
Structure and number response
$c = 0.04$
1.5
1.0
0.5
0.0
$S_{zz}(k)/2\bar{n}$
0
1
2
$k\sigma$
9
7
5
3
1
Γ
0.01
0.02
0.04
0.08
c